

MUZZARY BABAR

Software Engineer | Full-Stack · AI / ML · Python

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PROFESSIONAL SUMMARY

Computer Science graduate from GCU Lahore and software engineer who builds and ships real applications across full-stack web development and AI/ML. I designed and built a local-first RAG desktop app with a benchmark-driven retrieval pipeline and measured accuracy gains, and I designed, built, and deployed a full MERN web application end to end with an integrated AI assistant. I work daily with AI coding assistants, care about clean and tested code, and like measuring whether what I build actually works. Comfortable across Python, JavaScript, SQL, REST APIs, and both relational and NoSQL databases. I enjoy working independently, solving hard problems, and learning quickly in a collaborative team.

TECHNICAL SKILLS

Languages: Python, JavaScript (ES6+), SQL, C++, Java

AI / ML: Retrieval-Augmented Generation (RAG), information retrieval, vector search (pgvector), vector databases, embeddings (ONNX Runtime), retrieval evaluation (hit@k, recall@k, MRR), scikit-learn, Pandas, NumPy, NLP, prompt design, LLM APIs (OpenAI, Anthropic, Groq, Ollama, OpenRouter)

Frontend: React 18, React Router, JavaScript, HTML5, CSS3, Tailwind CSS, Axios, responsive single-page applications

Backend: Node.js, Express, FastAPI, REST API design, JWT authentication, middleware, server-side logic, third-party API integration

Databases: PostgreSQL (+ pgvector), MongoDB (Mongoose), MySQL (ER design, normalization, stored procedures), database design

Testing & Practices: Jest, Supertest, unit & integration testing, debugging, multi-dimensional code reviews (correctness, security, performance), clean and documented code, software architecture, security best practices

Tools, DevOps & Packaging: Git/GitHub, VS Code, Vercel, Railway, Linux, Docker (basics), PyInstaller, Inno Setup, ONNX Runtime; AI coding assistants (Claude Code, GitHub Copilot, Cursor); Agile/Scrum (coursework)

PROJECTS

CiteFinder, Local-First RAG Citation Assistant

Personal Project · 2026

Python · FastAPI · PostgreSQL/pgvector · ONNX · LLMs github.com/muzzary/citefinder

A local-first desktop app that answers questions only from a user's own PDFs, cites the exact file and page, and generates APA/Harvard/IEEE citations from confirmed metadata.

- Designed and built CiteFinder, a local-first retrieval-augmented generation (RAG) desktop app that answers questions strictly from a user's own PDF library and attributes every answer to a source file and page, shipped as a one-click Windows installer.
- Engineered a hybrid retrieval pipeline combining dense vector search (PostgreSQL + pgvector) with Postgres full-text search, fused via Reciprocal Rank Fusion, with adaptive multi-query expansion triggered only on weak results.
- Ran a benchmark-driven embedder evaluation on a 7,000+ chunk ground-truthed corpus; selecting and integrating e5-small-v2 (ONNX Runtime, CPU-only, torch-free) raised hybrid retrieval from 0.50 to 0.92 hit@1 and 0.72 to 0.94 MRR, reaching hit@5 1.00 and recall@5 0.86 on a labeled gold set, with thresholds tuned from data.
- Built a code-only citation engine producing APA, Harvard, and IEEE references (in-text and reference-list) for books, journal articles, and websites, with one-click DOI to CrossRef metadata auto-fill, plus a strict grounding and refusal gate so the model never fabricates sources.
- Packaged the entire stack as a self-contained desktop app (pywebview/WebView2, PyInstaller, Inno Setup) that bundles a portable PostgreSQL and an MSVC-built pgvector, with first-run database initialization, lifecycle management, and crash recovery, requiring no Docker or setup.
- Optimized CPU-only performance (model pre-warming, removed per-startup network calls) and made ingestion atomic; led a multi-dimensional code review (correctness, security, performance) that caught and fixed a multi-author citation bug.

JobTracker, Full-Stack MERN App with AI Assistant (deployed)

Personal Project · 2026

React · Node.js · Express · MongoDB · JWT · OpenRouter LLM · Jest job-tracker-muzzary.vercel.app · github.com/muzzary/job-tracker

- Designed, built, and deployed a full-stack MERN web application from scratch, helping job seekers track applications on a drag-and-drop Kanban board across five stages (Saved, Applied, Interview, Offer, Rejected), with the front-end on Vercel and the back-end on Railway.

- Built a REST API with 12 endpoints organized into clean route, controller, and model layers, following standard HTTP methods and status codes.
- Implemented JWT authentication with bcrypt-hashed passwords, protected routes with middleware, guarded login with rate limiting and Helmet security headers, and added server-side stale-session invalidation so deleted accounts are logged out promptly.
- Built an AI resume matcher: upload a PDF resume once, paste a job description on any card, and get a 0 to 100 match score plus a list of missing skills, with the LLM integrated entirely server-side (via OpenRouter) so the API key is never exposed, including model fallback and graceful rate-limit handling.
- Built and deployed an AI assistant feature that turns a job description into a tailored resume, a matching cover letter, and personalized interview prep, with results cached per job, a re-run option, and honest fit handling (it declines to over-tailor for a poor-match role rather than inventing skills).
- Wrote 20 automated API tests (Jest and Supertest) covering authentication and all job CRUD routes against a separate test database, and resolved real production-style issues including a DNS-level MongoDB connection failure and the stale-session security bug.
- Tech stack: React 18 with Vite, Node.js, Express, MongoDB Atlas, JWT, Tailwind CSS, Jest with Supertest.

Sales Automation System, Final Year Project

2025 to 2026

Python · n8n · GPT-4 · Puppeteer · Google Sheets · workflow orchestration

- Built a three-layer LinkedIn outreach automation pipeline that automates lead discovery, message personalisation, and CRM routing, reducing manual outreach effort by an estimated 80 percent.
- Used a language model for sentiment and skill-match analysis to score and prioritise leads before outreach, and designed fault-tolerant n8n workflow logic with retry queues and error-state handling for uninterrupted operation.

Student Grade Prediction, Machine Learning Model

2024

Python · scikit-learn · Pandas · Jupyter

- Trained and compared Decision Tree, SVM, and Random Forest models to predict student grades from academic and behavioural features.
- Improved accuracy through data cleaning, feature engineering, cross-validation, and hyperparameter tuning, and delivered a clean Jupyter Notebook with visualisations and a written performance analysis.

Cinema Database Management System

Academic Project

SQL · MySQL · ER design · normalization

- Designed and implemented a normalized relational database covering movies, bookings, users, and payments.
- Wrote optimized SQL queries and stored procedures, and delivered a full ER diagram and normalization documentation.

Online Learning Platform, System Design

Academic Project

Software Engineering · UML · DFD · SDLC

- Produced full pre-implementation documentation, including an SRS, use-case diagrams, DFDs, and system architecture for an e-learning platform designed to serve 500+ users, applying SDLC and agile planning principles.

EXPERIENCE

Independent Freelancer, Software, Web, ML & Database Projects

2023 to Present

- Delivered web, database, machine-learning, NLP, and software projects for international clients, working remotely from requirements through to delivery.
- Completed work for foreign university clients and UK-based clients, demonstrating strong written English communication, and maintained a 100% on-time delivery record with positive feedback across all engagements.
- Managed each project independently, making technical decisions and communicating clearly across time zones.

EDUCATION

BS Computer Science, Government College University (GCU), Lahore

2022 to 2026

Relevant coursework: Data Structures and Algorithms, Object-Oriented Programming, Machine Learning, Database Systems, Software Engineering, Computer Networks, Operating Systems, Web Development.

FSc Pre-Medical, Superior Group of Colleges, Lahore

2021 to 2022

Matriculation (Sciences), Qazi Group of Schools, Lahore

2019 to 2020

CERTIFICATIONS

- Claude Code 101, Anthropic Education (2026)

- Introduction to Claude Cowork, Anthropic Education (2026)
- Introduction to Model Context Protocol, Anthropic Education (2026)

ADDITIONAL

Languages: English (professional), Urdu (native)

Interests: AI and LLM applications, information retrieval, automation, open-source tools, system design, entrepreneurship, building and shipping side projects

Availability: Full-time roles, internships, and remote freelance work